



Research Paper

Physics-based two-scales reduced order model for real-time thermal management in immersion cooling data centers

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ABSTRACT

With the increasing power density in data center, liquid cooling data centers have attracted widespread attention and application. Thermal management has become a critical bottleneck restricting their further development. To address the thermal management challenges across multiple scales in data centers, this study established a multiscale model for the single-phase immersion cooling (SPIC) data centers and obtained the quantitative relationship between the thermal loading in the data center and the temperature increase of the chip. Then, a two-scales Proper Orthogonal Decomposition (POD)-based reduced order model (ROM) was developed for both the immersion tank and server scales, enabling rapid prediction of multiscale flow and temperature fields. Through boundary conditions delivery across the immersion tank and server scales, the rapid prediction of the thermal and flow fields in SPIC systems was achieved. Compared with CFD results, the maximum mean relative deviations of the POD-predicted velocity at the server inlets and outlets were 2.88% and 2.92%, respectively. For the internal three-dimensional temperature field of the server, the maximum mean relative deviation was 3.7%. The POD model achieved a computational speed up of approximately 1500 times over traditional CFD simulations.

1. Introduction

The widespread deployment of communication technologies and advances in cloud computing have significantly boosted data transfer rates, thereby drastically increasing the power consumption and heat generation of data centers [1]. The growing demand for parallel computing drives the thermal design power of microprocessors approaching 700 W, which is far beyond the air cooling capacity ~280 W [2], and thus the immersion cooling methods have attracted growing attention in recent years [3].

The superiority of single-phase immersion cooling (SPIC) over conventional cooling methods has also been demonstrated by many existing studies [4–6]. Specifically, Shah et al. [7] show that SPIC capacity is about three times that of air cooling for a cabinet. Notably, the immersion cooling systems can attain a Power Usage Effectiveness (PUE) as low as 1.07, in contrast to approximately 1.5 for typical air-cooled systems, indicating a roughly 36% reduction in power consumption.

Analysis on SPIC systems primarily relies on CFD simulation techniques, with a focus on analyzing the optimal configuration of key factors such as tank structure, heat sink design, dielectric fluid properties,

and operational parameters. Kim et al. [8] conducted a simulation study on SPIC using four 1 U sized printed circuit boards, and found that immersion cooling is the only method capable of maintaining component temperatures below the target temperature in an extreme condition. Mohammed et al. [9] performed CFD simulations on a high power density data center, and the results demonstrated that the immersion cooling system could reduce the maximum server temperature to 65 °C and improve the PUE from 2.0 in air-cooled systems to 1.01. Muneeshwaran et al. [10] focused on the inlet/outlet configuration and bypass effects in immersion cooling tanks, while Shrigondekar et al. [11] carried out a similar analysis with an emphasis on bypass behavior. Note that the uneven coolant distribution is a common issue in immersion cooling, particularly in server racks with high density. Such distribution non-uniformity can lead to insufficient heat dissipation in areas with lower flow rates, resulting in temperature inconsistencies within the rack. In this regard, Ni et al. [12] investigated immersion cooling across five printed circuit boards, examining how non-uniform coolant distribution in immersion cooling tanks may cause excessive temperature variations among servers.

The aforementioned studies are all based on immersion cooling

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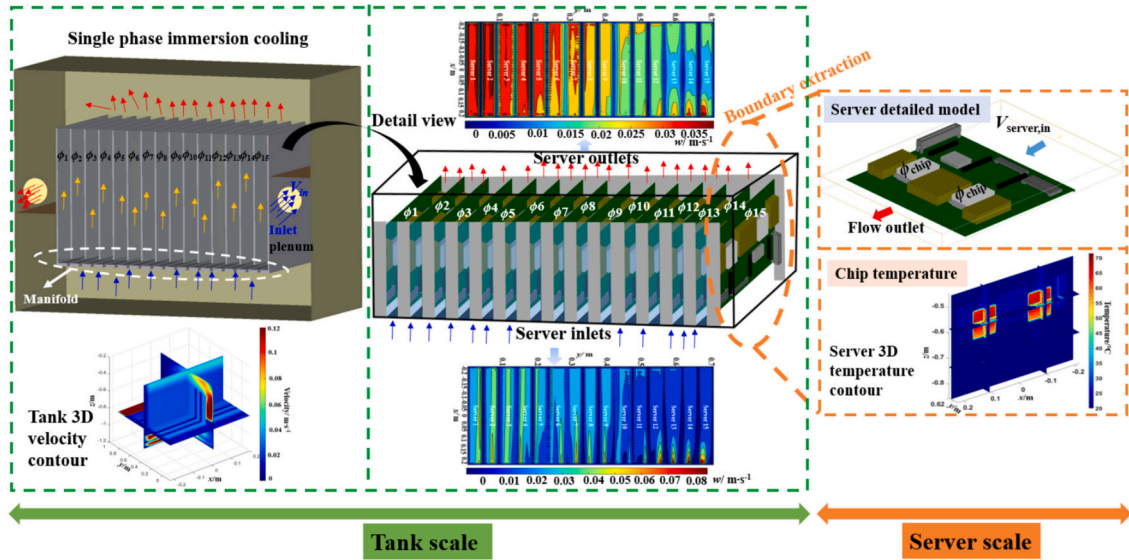


Fig. 1. Multiscale modeling of the SPIC data center.

research involving a small number of servers (fewer than five). However, in practical SPIC data centers, an immersion tank typically contains more than ten servers to achieve higher cooling power density [13,14]. These factors collectively hinder the direct extrapolation of conclusions from traditional small-scale server studies to real-world, large-scale application scenarios. Similar to multiscale models of air-cooled data centers [15–18], the cooling process in SPIC data centers also exhibits multiscale characteristics due to the wide range of length scales involved, which is generally divided into the chip scale, server scale, and immersion tank scale. Therefore, to reduce the cooling energy consumption of data centers, it is necessary to develop multiscale models that account for how heat generated by chips within the servers is removed by the cooling capacity of the immersion tank.

CFD simulations can be used to analyze the thermal environment of data centers and systematically study the effects of operating conditions [19]. However, they are not suitable for real time optimization and control applications due to their high computational cost and long execution time, which cannot satisfy the requirements of real-time control [20,21]. It is essential for the intelligent operations and maintenance in data centers by improving simulation speed and achieving rapid modeling [22,23]. Thus, various surrogate models have been developed in this regard. Among these, the reduced order model (ROM) based on the Proper Orthogonal Decomposition (POD) method is one of the most widely adopted techniques [24,25]. As discussed by our previous work [26], the POD-based ROMs can be successfully adopted to fast predict the fluid flow and thermal transport in data center, with sample data obtained either from CFD simulations or experimental measurements. Meanwhile, Samadiani and Joshi [27–29] demonstrate that the POD model is well-suited for turbulent flow modeling in data centers. Coincidentally, Jin et al. [30] employed the POD method combined with regression splines to predict the temperature distribution in a real data center. Dai et al. [31] conducted predictions on the air-cooled data center with limited temperature variables.

To the best of the authors' knowledge, existing multiscale and rapid prediction models have primarily focused on air-cooled data center systems, with limited attention to liquid-cooled scenarios. In SPIC systems, the coexistence of laminar and turbulent flows, combined with strong thermal interactions among densely packed servers, creates highly nonlinear coupling between fluid flow and heat transfer, resulting in extreme modeling complexity [32,33]. Capturing these dynamics in full-order CFD simulations is computationally prohibitive, and a single-level reduced-order model may struggle to accurately represent both large-scale system behavior and fine-scale local interactions.

Therefore, a two-scales POD-based reduced-order modeling approach becomes essential, as it can efficiently handle multiscale flow-thermal interactions while enabling rapid, accurate, and real time thermal management in data centers.

Therefore, this study constructs a two-scales POD-based ROM at both the tank and server scales based on a multiscale model of SPIC. Specifically, ROMs in each scale are developed for the immersion tank and the detailed three-dimensional (3D) server. By incorporating boundary condition transfer across scales, this approach effectively addresses the challenge of predicting chip temperatures under complex server power variations and spatially non-uniform cooling conditions, providing efficient thermal solutions for the design and deployment of next generation data center infrastructure.

2. Multiscale modeling and numerical model

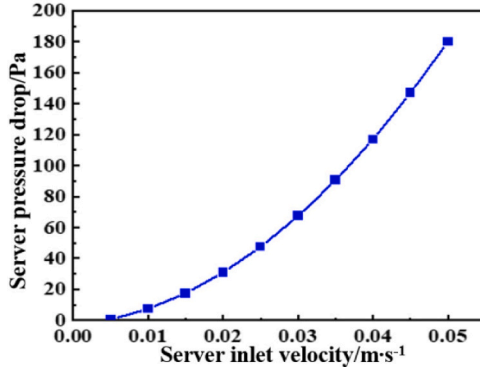
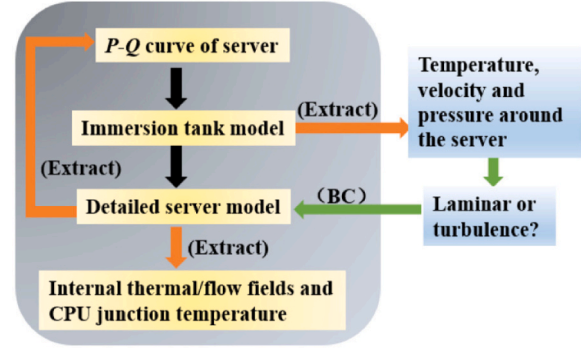
The cooling process in SPIC data centers is a typical multiscale phenomenon. In this study, two scales are considered: the immersion tank scale and the server scale. Fast prediction models are developed for each scale, and a two-scales ROM for SPIC is constructed through the transfer of boundary information between the involved scales. Specifically, the boundary conditions of the server model are extracted from the flow and temperature fields in the immersion tank.

2.1. Multiscale modeling of SPIC data center

The SPIC data center is primarily composed of immersion cooling tanks, servers, and the cooling system. Since each immersion tank operates independently and the heat is removed by the cooling water, the thermal management analysis is confined to the coupling between the immersion tank scale and the server scale.

As shown in Fig. 1, the immersion tank contains a total of 15 servers, each with a distinct power consumption denoted from ϕ_1 to ϕ_{15} . In terms of structural configuration, the tank incorporates a distribution chamber and a distribution manifold. The server model is simplified from the detailed Olympus server specification of the open compute project. The computational domain comprises two CPU heatsinks, twenty-four Dual In-line Memory Modules (DIMMs), and a platform controller hub component.

The multiscale numerical simulation method for SPIC data center is implemented as follows. First, a detailed server model is developed to obtain the relation between the pressure drop and flow resistance, which is then used as the input for the immersion tank model. In the tank scale

(a) P - Q curve of server

(b) Transfer of boundary conditions in multiscale modeling

Fig. 2. P - Q curve and boundary information transferred in multiscale model.

simulation, the servers are represented as porous media, while the server power consumption is imposed as heat sources within the tank. Subsequently, the boundary conditions extracted from the tank scale results are applied to the server scale simulations.

The immersion tank model consists of approximately 0.82 million computational cells, which is significantly less than a full-domain high-resolution simulation. By employing this multiscale simulation method, the computational cost can be significantly reduced.

2.2. Numerical simulation model description

2.2.1. Assumption and governing equation

The inlet flow states of the immersion cooling tank and the server are characterized using the Reynolds number [34], which is expressed as follows:

$$Re = \frac{\rho v D}{\mu} \quad (1)$$

where ρ , v and μ denote the density, inlet velocity, and dynamic viscosity of the coolant, respectively; D is the hydraulic diameter, which can be expressed as:

$$D = \frac{2bh}{b+h} \quad (2)$$

where b and h is the width and height of the rectangular flow passage, respectively. For the tank scale, $D = 0.04$ m. For the server scale, $b = 0.441$ m and $h = 0.042$ m.

The critical Re is related to the flow regime. In the viscous flow of the smooth tube, the critical Re is approximately 2300 [34]. Although taking $Re = 2300$ as the critical value between laminar and turbulent flow has certain limitations, for a specific actual structure, determining the turning point of laminar flow and turbulent flow requires the use of direct numerical simulation (DNS) methods. However, for the complex geometric structure in this paper, the computational load of DNS is extremely large and difficult to implement. Since the SST model can describe the transition behavior from laminar flow to turbulent flow relatively well [35], in the authors' previous studies, the SST turbulence model was employed to demonstrate the rationality of selecting $Re = 2300$ as the transition point between laminar and turbulent flow in the present work. Therefore, taking critical Re as 2300, either the turbulence model (k - ϵ) or the laminar flow model is selected accordingly.

The CFD simulations were performed using the commercial software ANSYS Icepak, which is based on the finite volume method (FVM). The numerical models for the internal flow fields of the tank and the server are developed under the following assumptions: (1) air is modeled as an incompressible fluid; and (2) the influence of the temperature field on the thermophysical properties of the fluid is neglected. The governing

equations for the laminar and turbulent flow and heat transfer at the steady state are presented in Eqs. (3) ~ (9), respectively.

Mass conservation equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (3)$$

Laminar momentum conservation equation:

$$u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} + f_i \quad (4)$$

Turbulence momentum conservation equation:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\eta + \eta_t) \frac{\partial u_i}{\partial x_j} \right] + S_i \quad (5)$$

where S_i is the source term and $S_i = \frac{\partial}{\partial x_j} \left[(\eta + \eta_t) \frac{\partial u_i}{\partial x_j} \right]$.

k -equation:

$$\rho \frac{\partial k}{\partial t} + \rho u_j \frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\eta + \frac{\eta_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + \eta_t \frac{\partial u_j}{\partial x_i} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \rho \epsilon \quad (6)$$

ϵ -equation:

$$\rho \frac{\partial \epsilon}{\partial t} + \rho u_j \frac{\partial \epsilon}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\eta + \frac{\eta_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + \frac{c_1 \epsilon}{k} \eta_t \frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - c_2 \rho \frac{\epsilon^2}{k} \quad (7)$$

Laminar energy conservation equation:

$$\rho c_p u_j \frac{\partial T}{\partial x_j} = \lambda \frac{\partial^2 T}{\partial x_j^2} + S_T \quad (8)$$

Turbulence momentum conservation equation:

$$\frac{\partial(\rho c_p T)}{\partial t} + \frac{\partial(\rho u_i c_p T)}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\lambda_{eff} \frac{\partial T}{\partial x_i} \right) + S_T \quad (9)$$

where ρ is the fluid density, u is the velocity, i and j are the spatial coordinates (usually taken as 1–3), ν is the kinematic viscosity, η is the dynamic viscosity, η_t is the turbulent dynamic viscosity (usually taken as $\eta_t = c_\mu \rho k^2 / \epsilon$), k is the turbulent kinetic energy, ϵ is the turbulent kinetic energy dissipation rate, $c_\mu = 0.09$, $\sigma_k = 1.0$, $\sigma_\epsilon = 1.3$, $\sigma_k = 1.0$, $c_1 = 1.44$ and $c_2 = 1.92$. λ is the thermal conductivity, λ_{eff} is the effective thermal conductivity ($\lambda_{eff} = \lambda + \lambda_t = \frac{\eta c_p}{Pr} + \frac{\eta_t c_p}{\sigma_T}$), Pr is the Prandtl number (usually taken as 0.85), $\sigma_T = 1.0$ and S_T is the heat source term.

2.2.2. Boundary condition

To obtain multiscale simulation results, geometric models were established separately for the immersion tank and the server. In the immersion tank simulations, the pressure-drop curve calculated from

the detailed server model was used as the input condition for the immersion tank porous media model. The server inlet velocity was set in the range of 0.005–0.050 m/s, with an interval of 0.005 m/s, yielding 10 sampling points; the variation of pressure drop with inlet velocity is shown in Fig. 2(a). The resulting pressure drop curve was then incorporated into the porous medium model of the immersion tank to represent the pressure loss of the fluid flowing through the server.

The boundary information transfer in the multiscale model of a single phase immersion cooling data center is illustrated in Fig. 2(b). First, a pressure drop resistance curve is obtained from the detailed server model and input into the immersion tank model. Subsequently, the server scale results calculated from the immersion chamber simulation are extracted and applied as boundary conditions to the server scale simulations, where either the laminar or turbulence model is selected according to the server inlet parameters. Finally, server scale simulations are performed to obtain the distributions of temperature and velocity inside the server, as well as the chip junction temperatures.

By constructing a multiscale model that couples the data center immersion tank and the server, computational resources can be significantly reduced, enabling rapid full field solutions while accurately capturing the influence of fluid distribution within the immersion chamber under different operating conditions on the chip junction temperatures of the servers.

2.3. Construction of sample database

2.3.1. Input variables

In the immersion tank model, a total of 16 variables are considered, including ϕ_1 to ϕ_{15} and the inlet velocity V_{in} . The detailed server model includes a total of 2 variables, namely the chip power consumption ϕ_{chip} and the inlet velocity of the server $V_{server,in}$.

The construction of sample database includes the immersion tank scale and server scale. Initially, the tank scale assumes that the power variables ($\phi_1 \sim \phi_{15}$) are increased from 300 W to 1050 W with a step of 250 W. Besides, the tank inlet velocity (V_{in}) is adjusted proportionally to the total power consumption of the immersion tank. Based on the full factorial design principle, a total of 4^{15} operating conditions would be required. It should be noted here that although 16 variables are involved, the value of one variable (V_{in}) is not independent, which is determined by the sum of the other 15 variables. Therefore, the total number of the operating conditions is 4^{15} . To simplify the sampling process while maintain data accuracy, a paired design approach is employed. Across the defined ranges of multiple variables, 37 sets of sample conditions are selected based on the paired design methodology. The paired design approach adopted in this study is based on pairwise testing, which is a combinatorial sampling strategy that focuses on covering all two-way parameter interactions rather than the full Cartesian product of all parameter levels. The fundamental idea behind pairwise testing is that most significant effects in a system arise from either single parameters or interactions between pairs of parameters, and higher order interactions tend to contribute relatively little to the overall system behavior in many practical cases. As a result, a design set that covers every possible combination of two parameters can achieve high coverage of the input space while dramatically reducing the number of required samples. This approach has been widely adopted and integrated into many test generation tools [30,36]. The number of samples was determined using the PICT (Pairwise Independent Combinatorial Testing) tool developed by Microsoft [37], which is a widely used implementation of pairwise testing. By applying PICT to the selected design variables and their respective levels, a total of 37 sample combinations were generated. This number represents the minimum set of samples required to ensure full pairwise coverage of the design variables. The corresponding values of each variable for these sets are detailed by Table S1 in the supporting information.

For the construction of the sample database at the server scale, the power consumption of chip, ϕ_{chip} , is increased from 150 W to 600 W

with a step of 90 W, and the inlet velocity, $V_{server,in}$, is set as 0.015, 0.03, and $0.045 \text{ m}\cdot\text{s}^{-1}$, respectively. The sample database for the server scale model comprises 18 operating conditions, as illustrated by Table S2 in the supporting information.

It is worth noting that the power ranges at different scales do not need to be directly correlated. For each scale, the selected power range represents typical and commonly reported power levels of racks and chips operating under single-phase immersion cooling in practical applications. Specifically, the tank scale range (300–1050 W) reflects the aggregated power levels of multiple heat sources, whereas the server scale range (150–600 W) corresponds to the power dissipation of individual server components or chips. Second, the choice of increment size is based on maintaining a reasonable number of discrete power levels for parametric analysis. The step sizes (250 W for the tank scale and 90 W for the server scale) were selected such that the power variable is represented by three to four levels, which provides sufficient resolution to capture the main trends while avoiding an excessive number of operating conditions.

2.3.2. Database generation

Firstly, the simulation of the immersion tank scale and server scale CFD models must be completed for 37 and 18 sets of sample conditions, respectively, and the output results must be stored to generate a sampling database. FC-40 fluorinated liquid is used as a single-phase immersion coolant. Reynold number (Re) is calculated based on the range of flow rates for the sample conditions, and a k - ϵ turbulence model is used for the CFD simulation at the immersion tank scale, while a laminar or turbulence model is selected for the server scale based on the inlet Re . In this study, the flow in the server was calculated to be laminar, and thus the laminar flow model was chosen to carry out the computational study as well as to establish a ROM.

For the thermal management of SPIC data center at the immersion tank scale, the stored data should contain the temperature (T), the velocity vector $U = (u, v, w)$, the turbulent fluctuation kinetic energy k , and the rate of dissipation of the turbulent fluctuation kinetic energy ϵ at each grid node. In order to obtain the temperature distribution of the model at the server scale, the sample database should contain the temperature T_{server} at each grid node under each sample condition.

In the process of obtaining the POD basis functions, $\mathbf{X}_T$, $\mathbf{X}_u$, $\mathbf{X}_v$, $\mathbf{X}_w$, $\mathbf{X}_k$, and $\mathbf{X}_\epsilon$ denote the values of various physical quantities in the immersion tank scale at different sample operating conditions and mesh nodes, respectively. $\mathbf{X}_{T,server}$ represents the temperature values at different sample operating conditions and mesh nodes at the server scale. These matrices are also referred as snapshot matrices ($\mathbf{X} \in \mathbb{R}^{N_c \times N}$, where N_c is the number of grid cells in the full-order system and N is the number of samples, with $N_c > N$). Taking $\mathbf{X}_T$ as an example, the matrix is expressed as shown in Eq. (10):

$$\mathbf{X}_T = \begin{bmatrix} T(1,1) & T(1,2) & \dots & T(1,N-1) & T(1,N) \\ T(2,1) & T(2,2) & \dots & T(2,N-1) & T(2,N) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ T(N_c-1,1) & T(N_c-1,2) & \dots & T(N_c-1,N-1) & T(N_c-1,N) \\ T(N_c,1) & T(N_c,2) & \dots & T(N_c,N-1) & T(N_c,N) \end{bmatrix} \quad (10)$$

where $N_c = 823,965$ and $N = 37$.

3. Construction of the POD-based ROM

3.1. Solving the POD basis function

POD is performed on the matrix $\mathbf{X}$ using the snapshot technique. The

Table 1The first 7 eigenvalues of sample data u and their energy distribution.

n	1	2	3	4	5	6	7
γ_n	108.49	0.11	0.05	0.03	0.02	0.01	0.01
ζ_n	99.76%	0.10%	0.05%	0.03%	0.02%	0.01%	0.01%
E_n	99.76%	99.86%	99.91%	99.94%	99.96%	99.97%	99.97%

extracted dominant modes enable the reconstruction of the temperature field as a linear combination of the corresponding basis functions. The number of retained modes is denoted by m , representing the truncation order. Thus, the temperature can be calculated by Eq. (11):

$$T(x) = c_1 \begin{bmatrix} \Phi_1(x_1) \\ \Phi_1(x_2) \\ \vdots \\ \Phi_1(x_{N_c}) \end{bmatrix} + c_2 \begin{bmatrix} \Phi_2(x_1) \\ \Phi_2(x_2) \\ \vdots \\ \Phi_2(x_{N_c}) \end{bmatrix} + \dots + c_m \begin{bmatrix} \Phi_m(x_1) \\ \Phi_m(x_2) \\ \vdots \\ \Phi_m(x_{N_c}) \end{bmatrix}, m \leq 37 \quad (11)$$

where Φ is the matrix of basis functions and c is the matrix of spectral coefficients.

As shown in Table 1 and Tables S3 to S9 in the supporting information, the selection of truncation orders for each physical quantity is based on the eigenvalues γ_n corresponding to each mode, which are arranged in the descending order. Here, ζ_n denotes the proportion of the n -th eigenvalue to the total sum of eigenvalues. The cumulative energy contribution E_n is computed, and an energy threshold of 99.5% is applied.

The extreme eigenvalue decay observed in Table 1 (e.g., $\gamma_1 = 108.49$ versus $\gamma_2 = 0.11$) indicates that the first POD mode captures the dominant physical behavior of the system. This behavior is consistently observed across different sample variables in this study, with the first mode generally possessing the highest energy content.

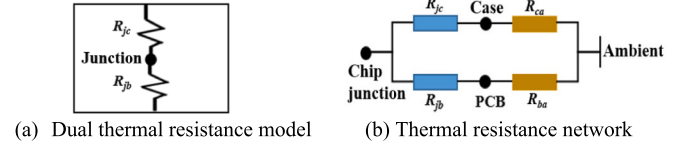
The truncation order m is then determined as the minimum number of modes required for E_n to reach the specified threshold. Based on this criterion, the truncation orders for the POD basis functions of the immersion tank model are selected $m = 1, m = 7, m = 7, m = 19, m = 1$ and $m = 1$ for u, v, w, T, k and ε , respectively. For the server POD model, an energy threshold of 99.9% is adopted, resulting in a truncation order of $m = 4$ for the temperature field.

For the tank scale analysis, an energy threshold of 99.5% is sufficient to capture the dominant flow and thermal features. At the server scale, however, using the same threshold of 99.5% would result in only a single POD mode being retained, which is insufficient to accurately represent the spatial details of the temperature distribution. To improve the prediction accuracy of the server scale temperature field while still keeping the number of retained modes limited, a higher energy threshold of 99.9% was therefore adopted. This choice allows additional meaningful modes to be included without significantly increasing the model complexity.

3.2. Solving the spectral coefficient in POD-based ROM

The methods for determining the spectral coefficients are classified into two categories: the Galerkin projection method and interpolation-based methods. The interpolation-based methods further include linear interpolation and spline interpolation. The POD method within the Galerkin projection framework exhibits notable generality and proves effective in addressing benchmark problems [38,39].

In the immersion tank model, the Galerkin projection method is used to compute the 3D temperature distribution. For the velocity field at the tank scale, linear interpolation is employed instead of Galerkin projection because only one varying parameter (inlet velocity) is involved. Under this condition, the dependence of velocity coefficients on the parameter is nearly linear, making linear interpolation sufficient and

**Fig. 3.** Chip dual thermal resistance model and network.

computationally more efficient. However, in the detailed server model, the traditional POD-Galerkin projection method exhibits significant limitations. These limitations primarily arise from the use of a dual thermal resistance model for CPU chips, in which the internal processes are not resolved by solving the governing equations.

The dual thermal resistance model is a lumped thermal model, as shown in Fig. 3(a). The dominant thermal paths include the junction to case thermal resistance R_{jc} and the junction to PCB thermal resistance R_{jb} . The corresponding thermal resistance network is illustrated in Fig. 3(b). R_{jc} and R_{jb} are obtained by chip manufacturers through standardized testing. In this study, $R_{jb} = 3.2 \text{ K} \cdot \text{W}^{-1}$ and $R_{jc} = 0.13 \text{ K} \cdot \text{W}^{-1}$ are adopted to model the dual thermal resistance chip package.

Because the dual thermal resistance model does not involve solving governing equations within the chip region and no spatial temperature or velocity fields are available there, the basis functions cannot be projected onto governing equations in the chip region. Therefore, applying projection based methods in this region is not feasible. A variable ϕ_{chip} defined as a volumetric heat source is applied directly to the chip's dual thermal resistance model. As a result, it is not feasible to project the basis functions onto the governing equations in the region ϕ_{chip} . In the rapid prediction of the temperature field in the server model, two variables simultaneously affect the temperature distribution. As a result, linear interpolation is insufficient to handle such scenario. In this study, the spline interpolation is employed to obtain spectral coefficients of the POD basic function at the server scale.

3.2.1. POD-Galerkin and linear interpolation for immersion tank ROM

The POD-Galerkin method reduces the system's dimensionality by projecting its governing equations onto the POD subspace, generating equations for the spectral coefficients c . The specific implementation procedure is as follows. First, the governing equation for energy in nondimensional form can be written as Eq. (12):

$$\left[\frac{\partial}{\partial x} \left(\lambda_{\text{eff}}(x, y, z) \frac{\partial \theta}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda_{\text{eff}}(x, y, z) \frac{\partial \theta}{\partial y} \right) + \frac{\partial}{\partial z} \left(\lambda_{\text{eff}}(x, y, z) \frac{\partial \theta}{\partial z} \right) \right] + S - \rho c_p \left(u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} + w \frac{\partial \theta}{\partial z} \right) = 0 \quad (12)$$

The temperature field is then expressed in the form of a series expansion, retaining only the first m terms of the series, as shown in Eq. (13):

$$\theta_M(x, y, z) = T_0(x, y, z) + \sum_{k=1}^m c_k \Phi_k(x, y, z) \quad (13)$$

Substituting Eq. (13) into Eq. (12), the residual of the governing equation is defined as R . By projecting R onto the first m basis functions and setting the projections to zero, Eq. (14) is obtained.

$$\begin{aligned}
0 = & \sum_{k=1}^m c_k [H_{ki}^x + H_{ki}^y + H_{ki}^z + E_{ki}^x + E_{ki}^y + E_{ki}^z - \rho c_p (M_{ki}^x + M_{ki}^y + M_{ki}^z)] \\
& + (T_i^{0x} + T_i^{0y} + T_i^{0z} + KT_i^{0x} + KT_i^{0y} + KT_i^{0z}) - \rho c_p (ST_i^{0x} + ST_i^{0y} + ST_i^{0z}) \\
& + \iiint_{\Omega} S \cdot \Phi_i(x, y, z) dx dy dz \quad (i = 1, 2, \dots, m)
\end{aligned} \quad (14)$$

Taking the x-direction as an example, where,

$$\begin{aligned}
H_{ki}^x &= \iiint_{\Omega} \lambda_{eff} \frac{\partial^2 \Phi_k}{\partial x^2} \Phi_i dx dy dz \quad KT_i^{0x} = \iiint_{\Omega} \frac{\partial \lambda_{eff}}{\partial x} \frac{\partial T_0}{\partial x} \Phi_i dx dy dz \\
E_{ki}^x &= \iiint_{\Omega} \frac{\partial \lambda_{eff}}{\partial x} \frac{\partial \Phi_k}{\partial x} \Phi_i dx dy dz \quad M_{ki}^x = \iiint_{\Omega} u \frac{\partial \Phi_k}{\partial x} \Phi_i dx dy dz \\
T_i^{0x} &= \iiint_{\Omega} \lambda_{eff} \frac{\partial^2 T_0}{\partial x^2} \Phi_i dx dy dz \quad ST_i^{0x} = \iiint_{\Omega} u \frac{\partial T_0}{\partial x} \Phi_i dx dy dz
\end{aligned} \quad (15)$$

By solving Eq. (14), the m unknown spectral coefficients c can be obtained. By applying this process identically to all 15 server regions, a system of $15 \times m_T$ equations is assembled. This system is then solved for the m_T coefficients using a least squares approach. A least squares formulation is adopted to obtain a consistent and optimal solution for the spectral coefficients across all regions.

The POD-based linear interpolation method involves decomposing the known snapshot velocity fields to acquire their spectral coefficients. The coefficients for an intermediate state are subsequently derived through the linear interpolation of these known coefficients. The complete velocity field is then reconstructed as a linear combination of the POD modes, weighted by the interpolated coefficients.

For the 3D steady-state convective heat transfer (turbulent flow) in the immersion tank, the Galerkin projection method is applied to each server region to obtain the tank-scale temperature spectral coefficients for the POD-based ROM, while the spectral coefficients in the ROM for the tank's velocity field are computed using a linear interpolation approach. The specific implementation details can be found in the authors' published works [26].

3.2.2. Spline interpolation method for the multivariate server ROM

The spectral coefficient matrix under the design condition is obtained by retaining the first m rows based on the truncation deviation. Consequently, the spectral coefficient matrices C_m for the multivariate server scale ROM is as shown below:

$$C_m = \begin{bmatrix} c_1^{(v_1, p_1)} & c_1^{(v_1, p_2)} & \dots & c_1^{(v_1, p_6)} & c_1^{(v_2, p_1)} & \dots & c_1^{(v_2, p_6)} & c_1^{(v_3, p_6)} \\ c_2^{(v_1, p_1)} & c_2^{(v_1, p_2)} & \dots & c_2^{(v_1, p_6)} & c_2^{(v_2, p_1)} & \dots & c_2^{(v_2, p_6)} & c_2^{(v_3, p_6)} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ c_m^{(v_1, p_1)} & c_m^{(v_1, p_2)} & \dots & c_m^{(v_1, p_6)} & c_m^{(v_2, p_1)} & \dots & c_m^{(v_2, p_6)} & c_m^{(v_3, p_6)} \end{bmatrix} \quad (16)$$

where the superscripts v and p represent the values of the sample variables $V_{server, in}$ and ϕ_{chip} , respectively.

In Eq. (16), the spectral coefficients c are denoted using a double index notation system, where the subscript indicates the order of the spectral coefficient. To obtain the spectral coefficients of each order under non-design conditions, a two-dimensional Cartesian coordinate system is constructed with $V_{server, in}$ as the horizontal axis and ϕ_{chip} as the vertical axis, as illustrated in Fig. S1. The variable v ranges from v_1 to v_3 , and p ranges from p_1 to p_6 .

This study aims to estimate the values of spectral coefficients for server models at arbitrary combinations of ($V_{server, in}$, ϕ_{chip}) using the tensor product B-spline interpolation method. Tensor product B-spline interpolation is a widely used numerical technique for multivariate function approximation [40,41]. Its core idea is to construct high dimensional interpolation functions through the tensor product of one-

Table 2

Variable parameters for the prediction cases.

	ϕ_1/W	ϕ_2/W	ϕ_3/W	ϕ_4/W	ϕ_5/W	ϕ_6/W	ϕ_7/W	ϕ_8/W
#1	500	550	950	720	320	650	900	1000
#2	310	310	310	570	570	770	1010	1040
#3	1030	100	1150	350	900	260	880	250

	ϕ_9/W	ϕ_{10}/W	ϕ_{11}/W	ϕ_{12}/W	ϕ_{13}/W	ϕ_{14}/W	ϕ_{15}/W	$V_{in}/m \cdot s^{-1}$
#1	920	950	520	890	670	420	910	-1.56
#2	1010	770	570	570	310	310	310	-1.25
#3	1090	479	986	370	1000	505	980	-1.48

dimensional B-spline basis functions. The specific steps are provided in the supporting information.

4. Results and discussions

4.1. Rapid predictions of inlet/outlet velocities for servers in immersion tank scale

To meet the demand for rapid prediction of the full field velocity distribution within an immersion cooling tank, this study employs a ROM approach. The method is based on POD, where the basis functions for velocity components u , v and w are extracted from snapshots, and the corresponding spectral coefficients are obtained using POD linear interpolation method.

To verify the effectiveness of this method, the proposed POD-based ROM is validated using CFD simulation results. Three representative prediction cases defined by the tank variable parameters $\phi_1 \sim \phi_{15}$ (server power consumption) and V_{in} (inlet velocity) are selected, with their values provided in Table 2.

Here, Test #1 represents a relatively random power distribution across servers; Test #2 corresponds to a power distribution where the middle servers exhibit higher power consumption than those on the sides, while Test #3 involves a configuration where the total power consumption of some servers slightly exceeds that of the training samples. To further investigate the influence of inlet velocity distribution in the cooling tank on the internal heat transfer characteristics of the servers, this study selects the server inlet cross section as a representative observation plane for analysis.

Fig. 4 illustrates the distributions of the z direction velocity component w at the server inlet sections under Test #1. Specifically, Fig. 4(a) presents the velocity fields predicted by the POD-based ROM; Fig. 4(b) shows the corresponding velocity field obtained from the full-order CFD simulation; and Fig. 4(c) displays the absolute deviation between the POD prediction and the CFD solution. A comparative analysis of the velocity field reveals that the inlet sections of Servers 1 through 9, located in the front region of the tank, exhibit a relatively uniform velocity distribution. The POD-based ROM predictions show good agreement with the full-order CFD results. The maximum absolute deviation occurs at the location of Server 14, with a value of $1.48 \times 10^{-3} m \cdot s^{-1}$. The average absolute deviation for this case is $1.53 \times 10^{-4} m \cdot s^{-1}$, and the mean relative deviation is 2.16%, which meets the accuracy requirements for engineering applications. These results indicate that the POD-based ROM can accurately capture the velocity distribution characteristics at the server inlet plane, thus providing a reliable foundation for subsequent analysis of heat transfer performance.

Fig. 5 illustrates the spatial distribution characteristics of the z direction velocity component (w) at the outlet sections of each server under Test #1 operating condition. A comparative analysis between the POD-based ROM predictions and the full-order CFD simulation results leads to the following conclusions: The outlet velocities of the front servers (Servers 1–6) in the immersion tank are relatively high, while

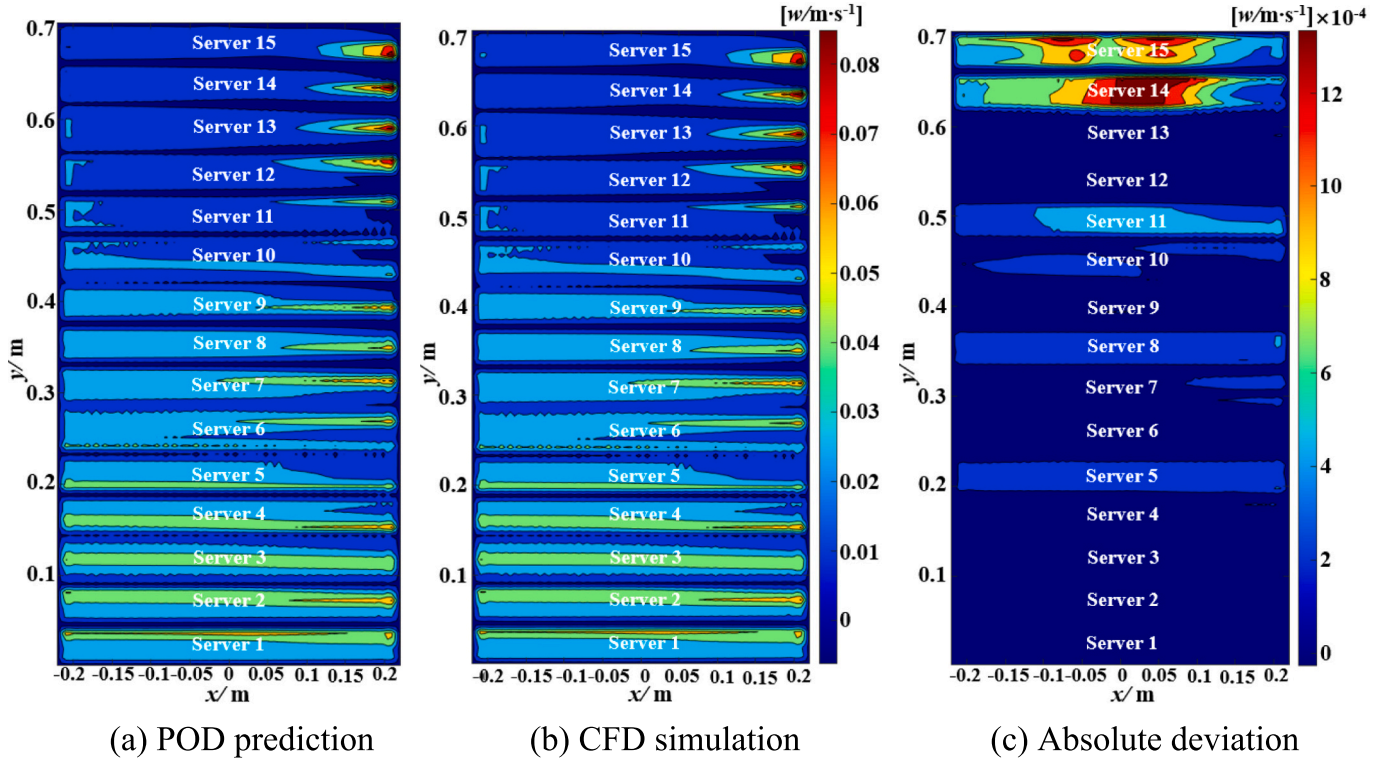


Fig. 4. Velocity contour of w at the server inlet sections under Test #1 prediction case.

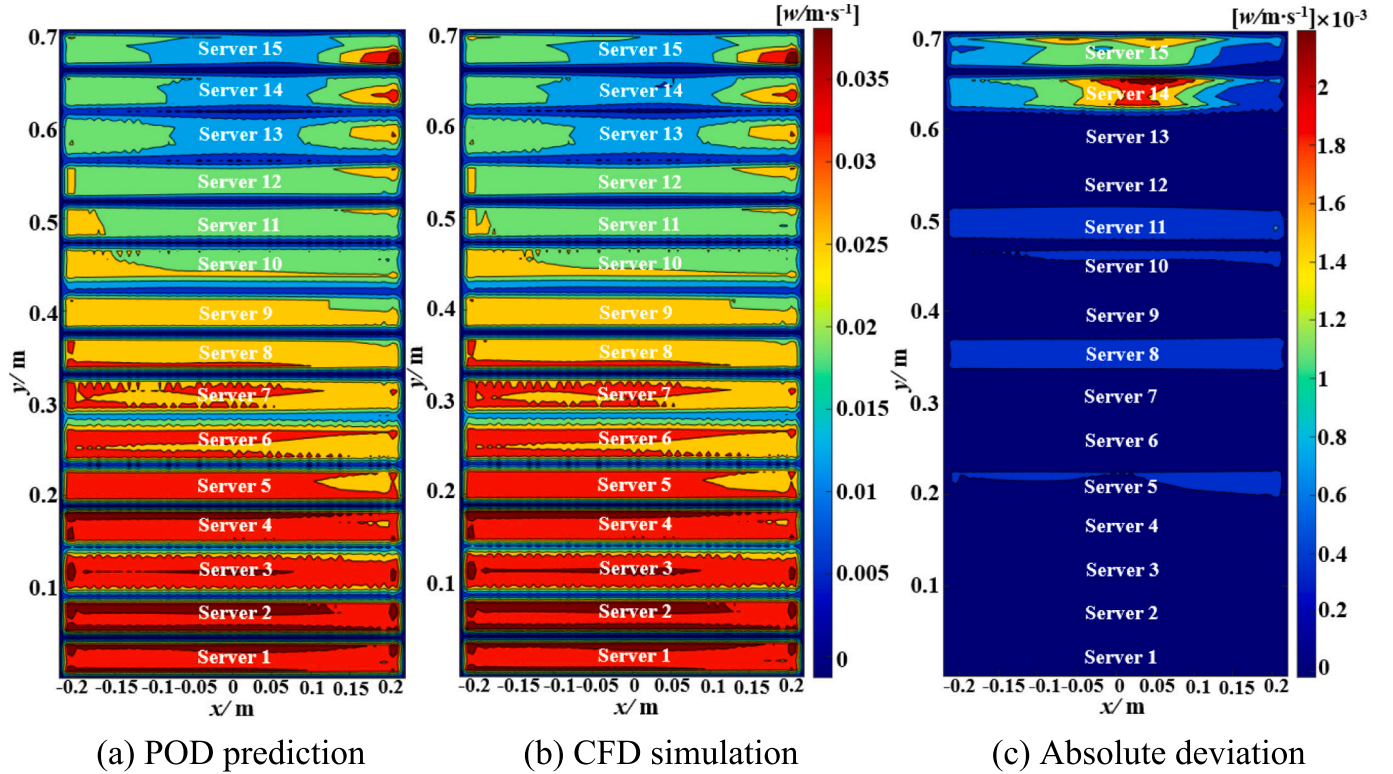


Fig. 5. Velocity contour of w at the server outlet sections under Test #1 prediction scenario.

those of the rear servers (Servers 7–15) are significantly lower. The velocity distribution predicted by the POD model generally agrees well with that obtained from the CFD simulations. Statistical analysis shows that the average absolute deviation is $1.94 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, with an

average relative deviation of 3.41%. The maximum absolute deviation $2.1 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, also occurs at the outlet of Server 14, located at the rear of tank. Examining the absolute deviation distributions at both the server inlets and outlets reveals that the regions with the largest

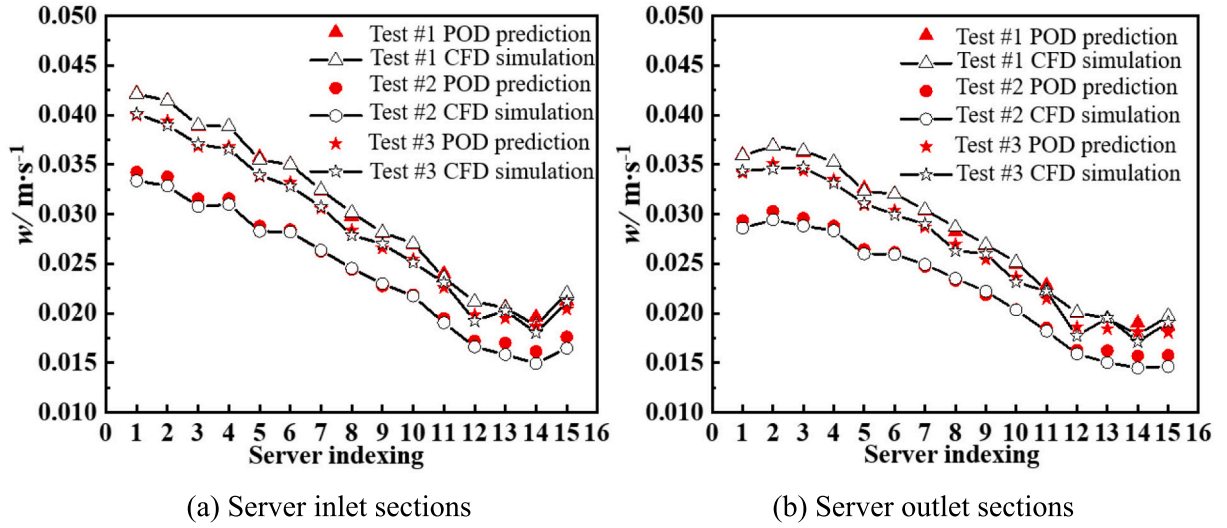
Fig. 6. Servers inlet/outlet w under different cases.

Table 3

Comparisons of POD and CFD prediction deviations for Test #1, Test #2 and Test #3.

Case	Servers inlet average $w/\text{m}\cdot\text{s}^{-1}$			Servers outlet average $w/\text{m}\cdot\text{s}^{-1}$		
	Maximum absolute deviation/ Server ID	Average absolute deviation	Average relative deviation	Maximum absolute deviation/ Server ID	Average absolute deviation	Average relative deviation
Test #1	0.00087/14	0.00023	0.96%	0.00119/14	0.000297	1.34%
Test #2	0.00077/13	0.00041	1.7%	0.00122/14	0.000568	2.92%
Test #3	0.00120/14	0.00060	2.88%	0.00109/13	0.000581	2.59%

deviations are concentrated around Server 14 and 15. These deviations are closely related to locally formed vortex structures in that region.

In practical data center thermal management, the prediction accuracy of ROMs under non-design operating conditions is of greater importance. Fig. 6 presents the average velocity distributions at the inlet and outlet cross sections of each server under three representative prediction cases (Test #1, Test #2 and Test #3). The comparison analyzes the agreement between the POD-based ROM predictions and the full-order CFD simulation results, with a truncation order $m = 7$.

From the average inlet velocities across the servers, it can be observed that in the immersion tank with a distributed chamber and manifold, the average velocity at both the inlet and outlet tends to decrease as the server position moves away from the outlet. A slight increase is noted after Server 12. Test #1 exhibits the highest average inlet and outlet velocities across all servers, while Test #3 shows the lowest since the inlet velocities for Test #1, Test #2 and Test #3 are $1.56 \text{ m}\cdot\text{s}^{-1}$, $1.25 \text{ m}\cdot\text{s}^{-1}$ and $1.48 \text{ m}\cdot\text{s}^{-1}$, respectively.

The deviations between the POD predicted and CFD simulated velocity component w at the inlet and outlet cross sections of each server under the three prediction cases are summarized in Table 3. For the inlet positions, under Test #1 condition, the average relative deviation reaches 0.96%, with the maximum absolute deviation of $8.7 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$ occurring at Server 14. In Test #2, the maximum absolute deviation is $7.7 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, located at Server 13, and the average relative deviation is 1.7%. For Test #3, the average relative deviation is 2.88%, with the maximum absolute deviation of $1.2 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$ also observed at Server 14.

Regarding the outlet positions, in Test #1, the maximum absolute deviation is $1.19 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, again at Server 14, with an average relative deviation of 1.34%. In Test #2, the largest deviation reaches $1.22 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$ at Server 14, accompanied by an average relative

deviation of 2.92%. Under Test #3, the average relative deviation is 2.59%, with the maximum absolute deviation of $1.09 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$ appearing at Server 13. These results indicate that the prediction errors tend to be more pronounced at the rear servers, particularly around Server 13 and 14.

The deviation values discussed above demonstrate that the POD-based ROM developed in this study is capable of accurately predicting flow behavior not only within the range of the sampled operating conditions but also for conditions that slightly exceed the sample range, such as Test #3. For critical cross sections, specifically, the server inlet and outlet planes, the average relative prediction error remains below 3%, which meets the required accuracy for reliable engineering prediction.

4.2. Rapid predictions of 3D velocity distributions in immersion tank scale

Using the developed POD-based ROM, fast predictions of the 3D velocity field within the entire immersion tank were performed for operating conditions Test #1 ~ #3. Velocity distributions plotted at the cross section located at $x = 0.008178 \text{ m}$, $y = 0.3535 \text{ m}$ and $z = -0.7829 \text{ m}$ are shown in Figs. 7(a ~ f).

From the velocity distributions, it can be observed that the fluorinated liquid enters the immersion tank through the inlet, experience a reduction in flow velocity as it passes through the server inlets, and then accelerates again near the tank outlet. Under all three prediction cases, the POD predicted velocity fields show excellent agreement with the CFD simulation results. The differences are negligible in the contour plots, with only a slight discrepancy observed at the outlet region of the immersion tank in Test #1 condition, where the liquid flows out after passing through the servers.

Figs. 7(a) and (b) present the 3D velocity field under Test #1

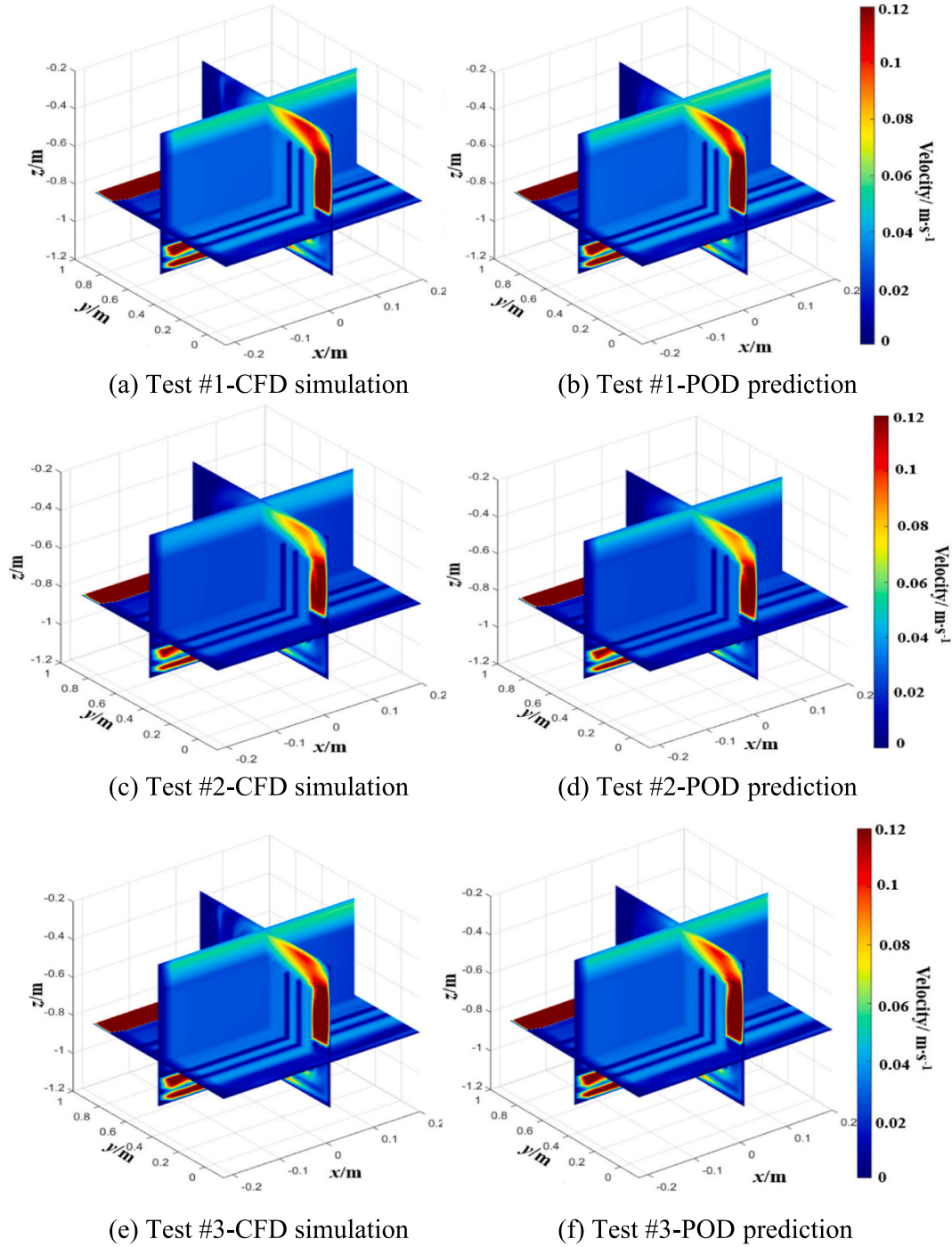


Fig. 7. Velocity contour of 3D flow fields under Test #1 ~ #3 cases.

prediction condition ($V_{in} = 1.56 \text{ m}\cdot\text{s}^{-1}$). It can be observed that the results obtained from the POD-based ROM are in excellent agreement with those from the tank scale CFD multiscale simulation. Specifically, for the x plane, the maximum absolute deviation is $0.02655 \text{ m}\cdot\text{s}^{-1}$, the average absolute deviation is $2.24 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average relative deviation is 1.58%. For the y plane, the maximum absolute deviation is $3.34 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, the average absolute deviation is $4.95 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average relative deviation is 1.26%. For the z plane, the maximum absolute deviation is $1.49 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, the average absolute deviation is $2.5 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average relative deviation is 1.44%.

Figs. 7(c) and (d) show the velocity field prediction results under Test #2 operating condition ($V_{in} = 1.25 \text{ m}\cdot\text{s}^{-1}$), comparing the fast

prediction from the POD-based ROM with the full-order CFD simulation. In the x plane, the maximum deviation is $0.014078 \text{ m}\cdot\text{s}^{-1}$, and the average absolute deviation is $5.39 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 1.96%. In the y plane, the maximum deviation is $0.00668 \text{ m}\cdot\text{s}^{-1}$, and the average absolute deviation is $4.54 \times 10^{-3} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 1.92%. In the z plane, the maximum deviation is $0.0021 \text{ m}\cdot\text{s}^{-1}$, and the average absolute deviation is $4.83 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 1.84%.

Figs. 7(e) and (f) present a comparison between the POD predicted results and CFD simulated results under Test #3 operating condition ($V_{in} = 1.48 \text{ m}\cdot\text{s}^{-1}$). The POD-based ROM once again yields results that closely match those of the full-order CFD simulation. For the x plane, the maximum deviation is $2.92 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average absolute

Table 4

Comparisons of absolute deviations for the entire immersion tank velocity between POD and CFD for Test #1 ~ #3.

Decision criteria	Proportion of Test #1	Proportion of Test #2	Proportion of Test #3
Absolute deviation $\geq 0.01 \text{ m}\cdot\text{s}^{-1}$	0.981%	2.05%	0.78%
$0.01 \text{ m}\cdot\text{s}^{-1} > \text{Absolute deviation} \geq 0.005 \text{ m}\cdot\text{s}^{-1}$	1.87%	6.95%	2.19%
$0.005 \text{ m}\cdot\text{s}^{-1} > \text{Absolute deviation} \geq 0.001 \text{ m}\cdot\text{s}^{-1}$	10.04%	31.74%	8.82%
Absolute deviation $< 0.001 \text{ m}\cdot\text{s}^{-1}$	87.11%	59.26%	88.21%

deviation is $8.78 \times 10^{-5} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 0.16%. In the y plane, the maximum deviation is $2.6 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average absolute deviation is $1.34 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 0.074%. Besides, in the z plane, the maximum deviation is $2.91 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, and the average absolute deviation is $1.2 \times 10^{-4} \text{ m}\cdot\text{s}^{-1}$, while the average relative deviation is 0.385%.

In those cases, the entire immersion tank domain contains a total of 823,965 mesh nodes. The prediction deviations between CFD simulation and POD results under each test condition is summarized in Table 4.

As shown in the Table 4, the lowest proportion of nodes with an absolute deviation less than $0.001 \text{ m}\cdot\text{s}^{-1}$ occurs under Test #2 condition, which corresponds to the lowest inlet velocity among three cases. In contrast, Test #3 exhibits the highest proportion of nodes with absolute deviation below $0.001 \text{ m}\cdot\text{s}^{-1}$, reaching 88.21%. For Test #1, this proportion also reaches 87.11%. These results indicate that the

developed ROM is capable of maintaining high prediction accuracy across the vast majority of data points throughout the domain.

By extracting the dominant flow field modal features, a ROM was constructed to enable rapid prediction of the full field velocity distribution. To evaluate the computational efficiency of the POD-based ROM, performance tests were conducted on a workstation equipped with a 2.99 GHz CPU and 256 GB of RAM. For the simulation of 100 new operating conditions, the traditional CFD approach requires approximately 216.67 h to complete all cases. In contrast, the POD-based ROM only takes 3 min to construct the basis functions, and 6 s to solve for the spectral coefficients. Predicting 100 velocity components takes 13 min (3 min + $100 \times 6 \text{ s}$), meaning the total time required to predict the 3D velocity field is only 39 min. Compared with the CFD model, the POD method achieves a speedup of approximately 333.3 times. This substantial improvement in efficiency is primarily due to the effective dimensionality reduction achieved by modal truncation in the POD method, which retains the essential physical characteristics of the flow field. It is noteworthy that despite the significantly reduced computational cost, the prediction accuracy of the ROM remains within an acceptable range, with the relative deviation from the CFD results below 2%, thus meeting the precision requirements for engineering applications.

4.3. Rapid predictions of 3D temperature distributions in immersion tank scale

In the temperature field prediction, the truncation order of the ROM is 19. The POD-Galerkin projection method is employed to construct projection equations for the heat source regions (i.e., the 15 server

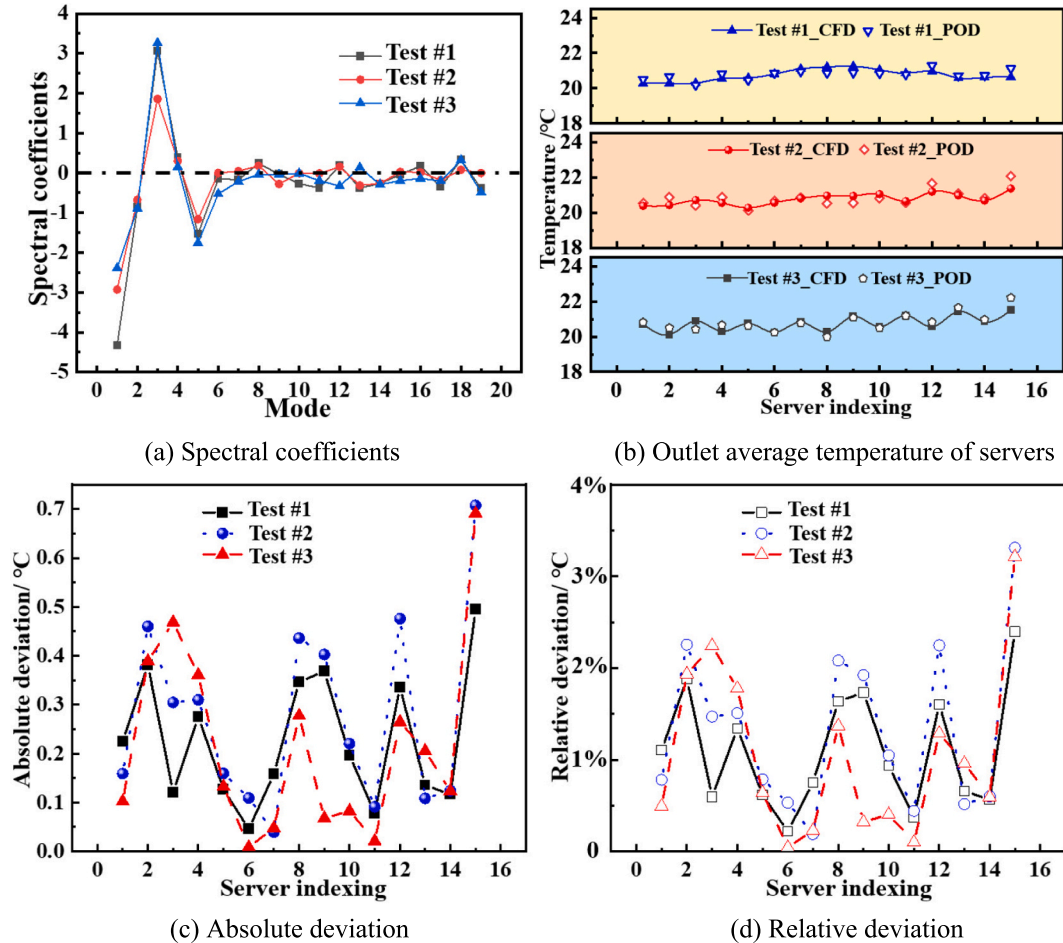


Fig. 8. Temperature prediction of the immersion tank at server outlets under Test #1 ~ #3 conditions.

Table 5
Power distribution among server components.

Component	Quantity	Power consumption per unit /W	Total power consumption of devices /W
ϕ_{chip}	2	425	850
PCH	1	30	30
DIMMs	24	5	120
Server Power	/	/	1000

regions). Under this framework, a total of $19 \times 15 + 1 = 286$ equations (including one equation representing the constant inlet temperature region) are established, and the coefficients of 19 POD basis functions are determined by solving this system using the least squares method. As shown in Fig. 8 (a), the coefficients of basis functions for three Test cases decrease as the modal index increases, indicating that the first few modes dominate the thermal field behavior. Since the inlet temperature of the tank is assumed to be constant at 20 °C for all conditions, the temperatures at the server inlets are almost identical, with minor variations. Fig. 8(b) presents the average outlet temperature of each server under three prediction conditions (Test #1, Test #2 and Test #3). The predicted average outlet temperatures from the POD model closely match the CFD simulation results. It is evident that differences in heat generation lead to variations in outlet temperatures among the servers. Specifically, in Test #3, where the server power consumption alternates between high and low values, the outlet temperatures of the servers exhibit a corresponding alternating pattern, reflecting the influence of non-uniform thermal loading on the temperature distribution.

Figs. 8(c) and (d) respectively illustrate the absolute and relative deviations between the POD predicted and CFD simulated average outlet temperatures for each server under three prediction cases. As shown in the Figs. 8(a)–(d), both the maximum absolute deviation and the maximum relative deviation occur at Server 15. Across the three test cases, the maximum absolute deviation reaches 0.71 °C, while the maximum relative deviation is 3.31%. These results indicate that the ROM provides temperature predictions that meet the accuracy requirements for engineering applications under all three operating conditions.

Here, the computational efficiency of the POD-based ROM for predicting the 3D temperature field at the immersion tank scale is evaluated. Using the POD model, the prediction of 100 operating conditions is completed within 13 min, including 3 min for computing the basis functions and 600 s for solving the spectral coefficients for all 100 test

cases. In contrast, the full-order CFD model requires 216.67 h to compute the same set of operating conditions. This demonstrates that the POD approach achieves a computational speed up of 1000 times compared with the CFD simulations.

4.4. Rapid predictions of 3D temperature distributions in server scale

A multiscale modeling approach combining the immersion tank model with a POD-based ROM is employed to enable rapid prediction of the temperature field at the server scale. Taking Server 8 under Test #1 condition, characterized by the highest total power consumption ($\phi_8 = 1000$ W), as an example, the predicted tank scale results (see Fig. 4(a)) indicate that the average inlet fluid velocity for this server is $0.0298 \text{ m}\cdot\text{s}^{-1}$, i.e., $V_{\text{server,in}} = 0.0298 \text{ m}\cdot\text{s}^{-1}$. The power distribution among the server components is listed in Table 5, where the chip power consumption is $\phi_{\text{chip}} = 425$ W.

For the case ($V_{\text{server,in}}, \phi_{\text{chip}}$) = (0.0298 $\text{m}\cdot\text{s}^{-1}$, 425 W), the spline interpolation is performed to obtain the corresponding spectral coefficients. The predicted 3D temperature field inside the server is shown in Fig. 9(a), while the CFD simulation results are presented in Fig. 9(b) for reference. Under this condition, the highest internal server temperature appears at the chip region, reaching 75.81 °C. According to the statistical analysis over the entire computational domain (1,297,863 nodes), the POD model yields an average absolute deviation of 0.61 °C and an average relative deviation of 1.18% compared with CFD results, both of which meet the design accuracy requirements.

To investigate whether the chip surface temperature is higher in the server with the lowest inlet velocity in the server or with the highest chip power consumption, a rapid 3D temperature field prediction was also carried out for Server 14 under Test #1 condition, which has the lowest average inlet velocity ($V_{\text{server,in}} = 0.01877 \text{ m}\cdot\text{s}^{-1}$), and the results are shown in Fig. 10. According to Table 2, the total server power is 420 W. By subtracting the power of other components (listed in Table 5) from this total, the chip power consumption ϕ_{chip} is thus determined to be 135 W.

Accordingly, spectral coefficient interpolation is performed for the condition ($V_{\text{server,in}}, \phi_{\text{chip}}$) = (0.01877 $\text{m}\cdot\text{s}^{-1}$, 135 W) to obtain the corresponding coefficient values for this prediction case. Fig. 10 presents the predicted 3D temperature field and the CFD simulation result. The average absolute deviation of the temperature across the domain is 0.16 °C, with 98.4% of grid nodes exhibiting absolute temperature deviations less than 1 °C. The average relative deviation is 0.65%, and the maximum temperature on the chip surface reaches 39.11 °C.

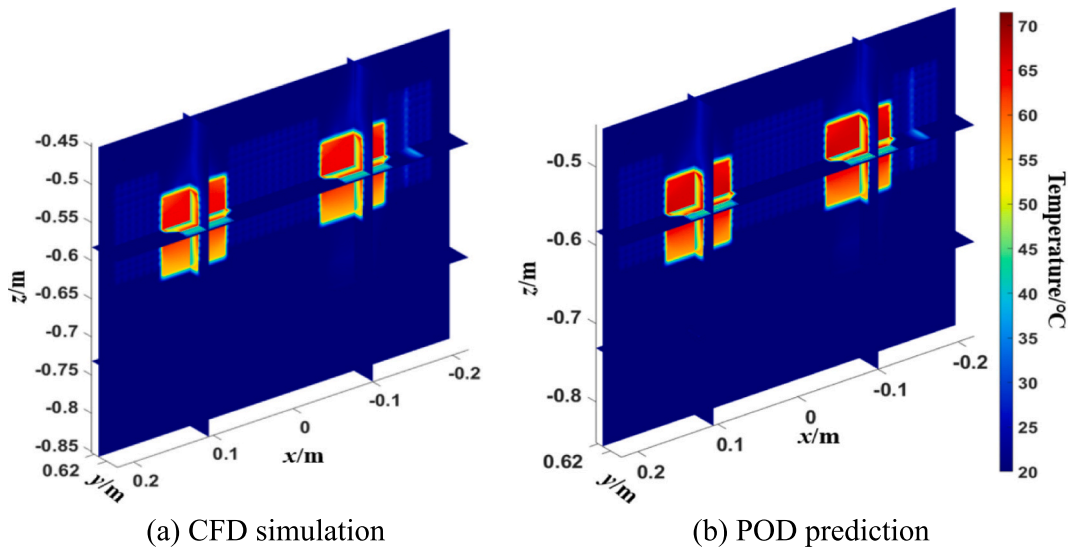


Fig. 9. Temperature contour of Server 8 (highest power consumption) in Test #1.

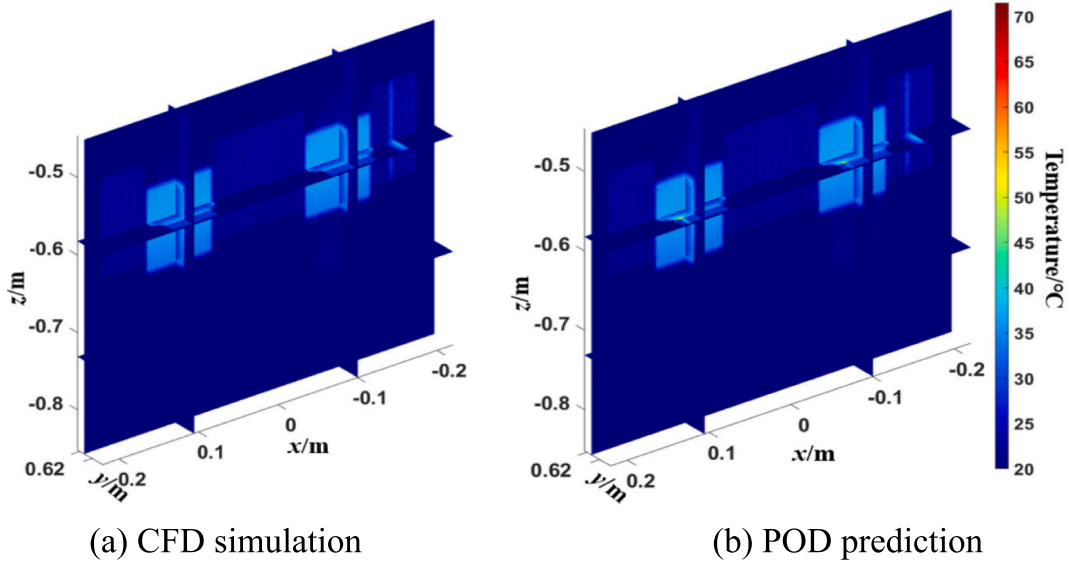


Fig. 10. Temperature contours of Server 14 (lowest inlet velocity) in Test #1.

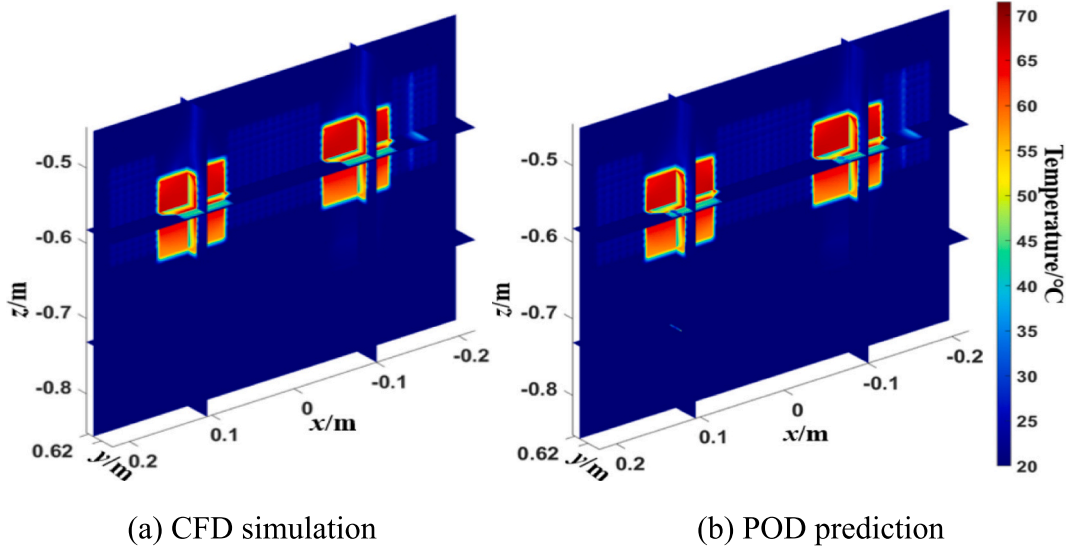


Fig. 11. Temperature contours of Server 8 (highest power consumption) in Test #2.

Similarly, under Test #2 conditions, rapid temperature field predictions are conducted for both the server with the highest total power consumption and the server with the lowest inlet velocity. The corresponding prediction cases are defined as Server 8 with $(V_{\text{server,in}}, \phi_{\text{chip}}) = (0.02454 \text{ m}\cdot\text{s}^{-1}, 445 \text{ W})$, and Server 14 with $(V_{\text{server,in}}, \phi_{\text{chip}}) = (0.01494 \text{ m}\cdot\text{s}^{-1}, 80 \text{ W})$. The predicted temperature distributions for these two servers are shown in Figs. 11 and 12, respectively. For Server 8, as shown in Fig. 11, the average absolute error of the overall temperature field is 0.20°C , with 97.8% of grid nodes below 1°C . The average relative error is 0.7%, and the chip surface temperature is 76.17°C .

For Server 14, as shown in Fig. 12, the average absolute deviation of the overall temperature field is 0.87°C , with 96.6% of grid nodes below 1°C . The average relative deviation is 3.7%, and the chip surface temperature is 32.02°C .

Predictions of the rapid temperature field were obtained for the server with the highest total power consumption and the server with the lowest inlet velocity, respectively, under the conditions of Test #3. The corresponding prediction cases are Server 3 with $(V_{\text{server,in}}, \phi_{\text{chip}}) =$

$(0.03705 \text{ m}\cdot\text{s}^{-1}, 500 \text{ W})$ and Server 14 with $(V_{\text{server,in}}, \phi_{\text{chip}}) = (0.01804 \text{ m}\cdot\text{s}^{-1}, 177.5 \text{ W})$. The temperature distributions for these two servers under the given conditions are shown in Figs. 13 and 14, respectively.

For Server 3, as shown in Fig. 13, the average absolute deviation of the overall temperature field is 0.072°C , with 99.02% of the grid nodes exhibiting an absolute temperature deviation less than 1°C . The average relative deviation is 0.24%, and the chip surface temperature is 80.23°C .

For Server 14, as shown in Fig. 14, the average absolute deviation of the overall temperature field is 0.092°C , with 98.6% of the grid nodes exhibiting an absolute temperature deviation less than 1°C . The average relative deviation is 0.37%, and the chip surface temperature is 44.73°C .

4.5. Summary of prediction results

By utilizing the constructed two-scales ROM, the average inlet velocity of each server can be obtained from the tank-scale ROM model. Then, through the multiscale model, the server parameters are input into

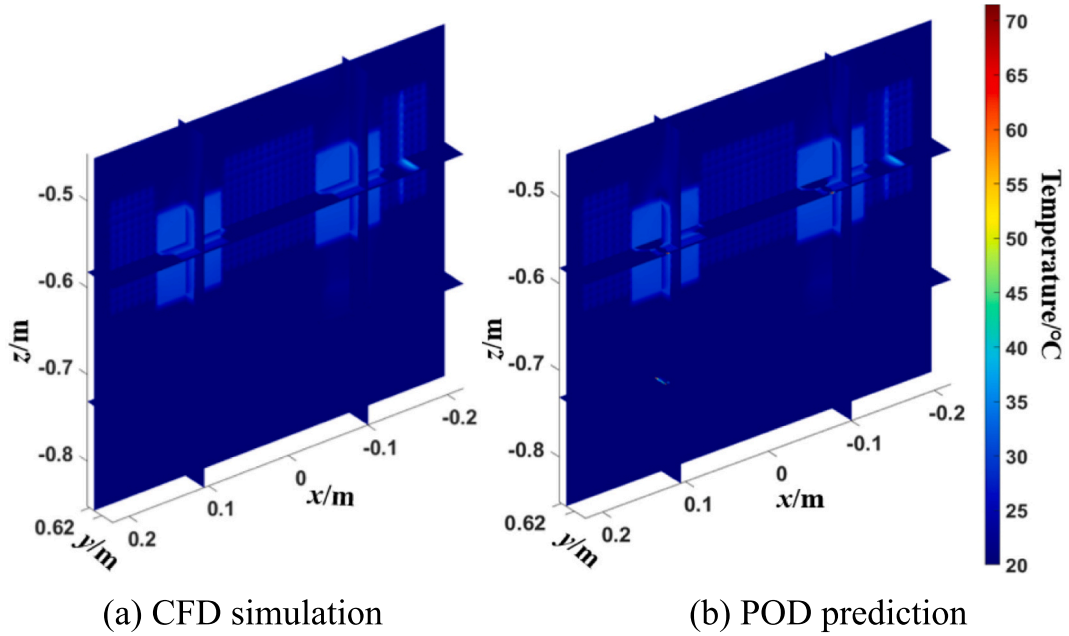


Fig. 12. Temperature contours of Server 14 (lowest inlet velocity) in Test #2.

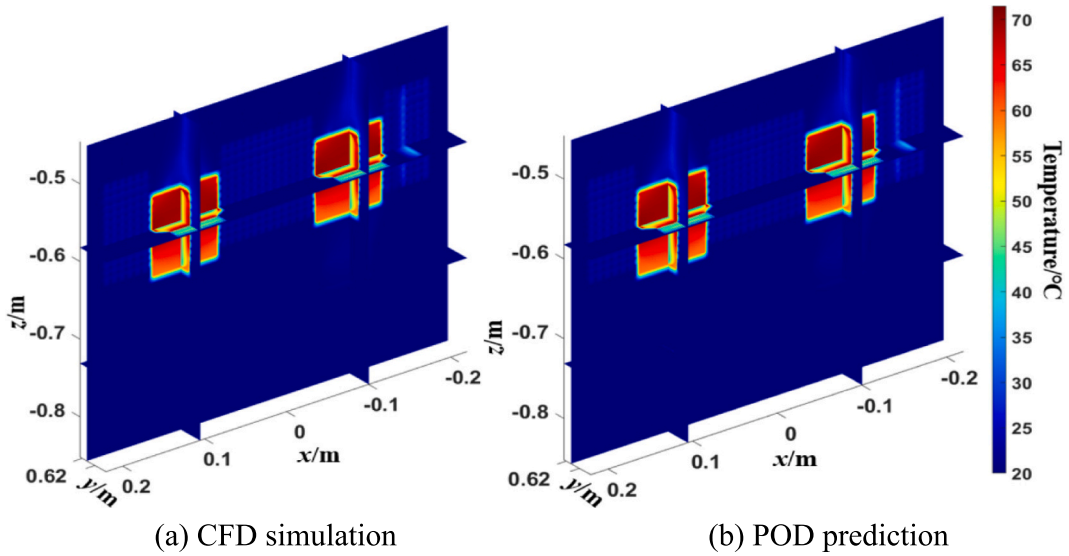


Fig. 13. Temperature contours of Server 3 (highest power consumption) in Test #3.

the server scale ROM to predict the 3D temperature distribution within the servers. Although the tank inlet parameters are identical, the prediction results for three randomly selected operating conditions (Table 6) reveal significant chip temperature differences, attributable to variations in server power consumption and average server inlet velocity.

Under these conditions, for Test #1, the maximum chip surface temperature difference between the server with the lowest inlet velocity and the server with the highest power consumption reaches $36.7\text{ }^{\circ}\text{C}$; for Test #2, the difference is $44.15\text{ }^{\circ}\text{C}$; while for Test #3, it is $35.5\text{ }^{\circ}\text{C}$. Therefore, greater attention should be paid to servers with higher power consumption, as they correspond to higher chip surface temperatures.

Fig. 15 presents the chip temperatures under the three predicted operating conditions by comparing two extreme cases: the server with the highest power consumption and the server with the lowest inlet velocity. Among the three prediction cases, the highest chip temperature occurs in Server 3 under Test #3 condition, with a surface temperature

reaching as high as $80.23\text{ }^{\circ}\text{C}$. This condition corresponds to a total tank power consumption of 10.33 kW and an inlet flow velocity of $1.48\text{ m}\cdot\text{s}^{-1}$. Under this scenario, the maximum temperature of all server chips remains below $85\text{ }^{\circ}\text{C}$, meeting the operational requirements.

To quantitatively demonstrate the computational efficiency advantages of the POD-based ROM within the multiscale modeling framework, Fig. 16 presents a comparative analysis highlighting the significant difference in computational time between the POD approach and conventional CFD simulations. For the prediction of server chip temperatures under 100 new operating conditions using a multiscale CFD model, the CFD based simulation requires approximately 216.67 h to complete the immersion tank scale computation. Subsequently, based on the boundary conditions obtained from the tank scale, the server scale CFD simulations take an additional 250 h , resulting in a total computational time of 466.67 h ($216.67\text{ h} + 250\text{ h}$). In contrast, the POD-based ROM approach requires only 15.67 min ($4\text{ min} + 100 \times 7\text{ s}$) to predict chip temperatures at the server scale. Specifically, 4 min are used for the

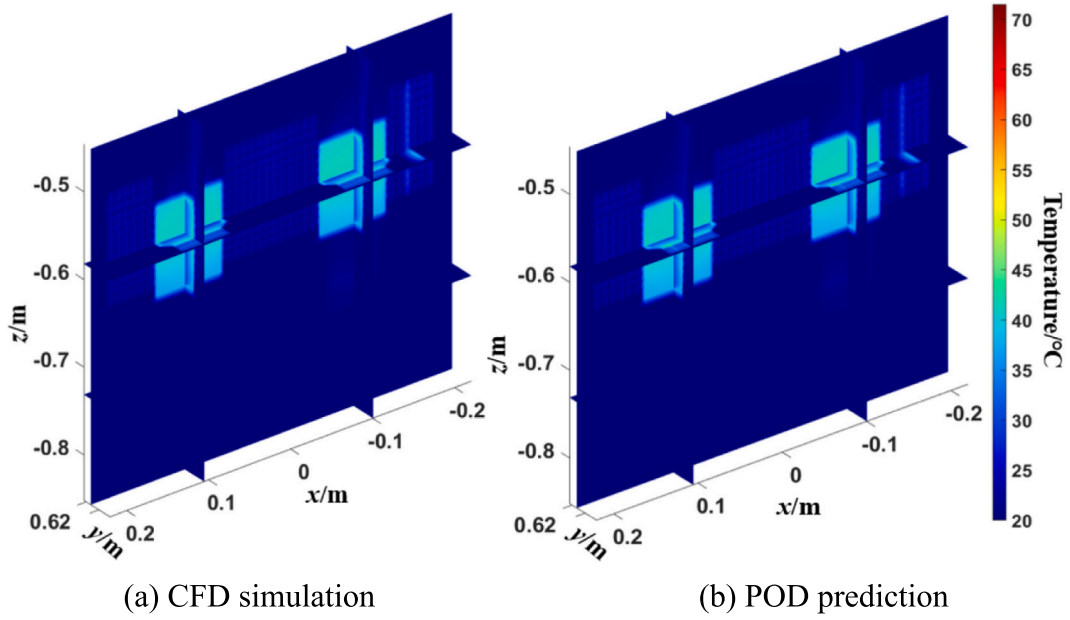


Fig. 14. Temperature contours of Server 14 (lowest inlet velocity) in Test #3.

Table 6

Temperature and velocity of tank and servers under three different prediction conditions.

Test case	Prediction condition parameters		Tank ROM		Server ROM
	Inlet velocity of the tank V_{in} / $m \cdot s^{-1}$	Total power consumption of the tank ϕ_{total} / kW	Server power consumption ϕ_{server} / kW	Average server inlet velocity / $m \cdot s^{-1}$	Maximum temperature of the chip / $^{\circ}C$
Test #1	1.56	10.87	1	0.0298	75.81
Test #2	1.25	8.74	0.42	0.01877	39.11
			1.04	0.02454	76.17
Test #3	1.48	10.33	0.31	0.01494	32.02
			1.15	0.03705	80.23
			0.505	0.01804	44.73

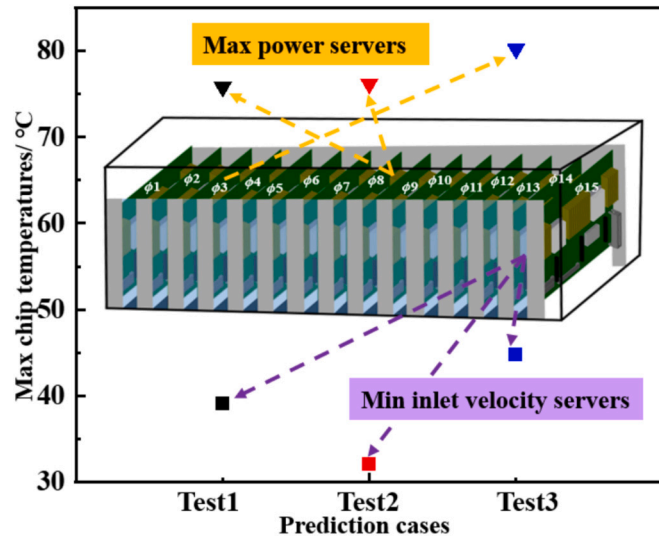


Fig. 15. Predicted maximum chip surface temperatures under Test #1 ~ #3 conditions.

generation of basis functions, and 700 s per case for the computation of spectral coefficients. Including the immersion tank POD prediction, which takes 13 min, the total computational time amounts only 28.67 min. This demonstrates that the POD-based ROM achieves

approximately a 1500 times improvement in computational efficiency compared with CFD simulations for multiscale SPIC applications. Once the POD basis functions are constructed offline, predicting a new operating condition only requires solving for the spectral coefficients. Specifically, the online prediction time consists of approximately 6 s for the tank scale ROM and 7 s for the server scale ROM, resulting in a total of about 13 s per operating condition for predicting the chip temperature distribution. This computational cost is well within the requirements of real time thermal management.

5. Conclusion

In this study, a two-scales POD-based ROM framework was developed for single-phase immersion cooling (SPIC) in data centers, comprising a tank scale and a server scale model. By leveraging boundary condition delivery within a multiscale modeling approach, rapid multiscale prediction of the SPIC system was achieved. At the immersion tank scale, a ROM incorporating 15 server power variables and an inlet flow velocity variable was constructed. The 3D velocity field was modeled using the POD linear interpolation method, while the temperature field was captured via the POD-Galerkin projection approach. At the server scale, a bivariate POD-spline interpolation ROM was established, enabling rapid prediction of both the 3D internal temperature distribution and the chip surface temperature for each server. The main conclusions obtained are as follows:

- (1) First, the 3D velocity field and the temperature field within the immersion tank were predicted. The results demonstrate that the

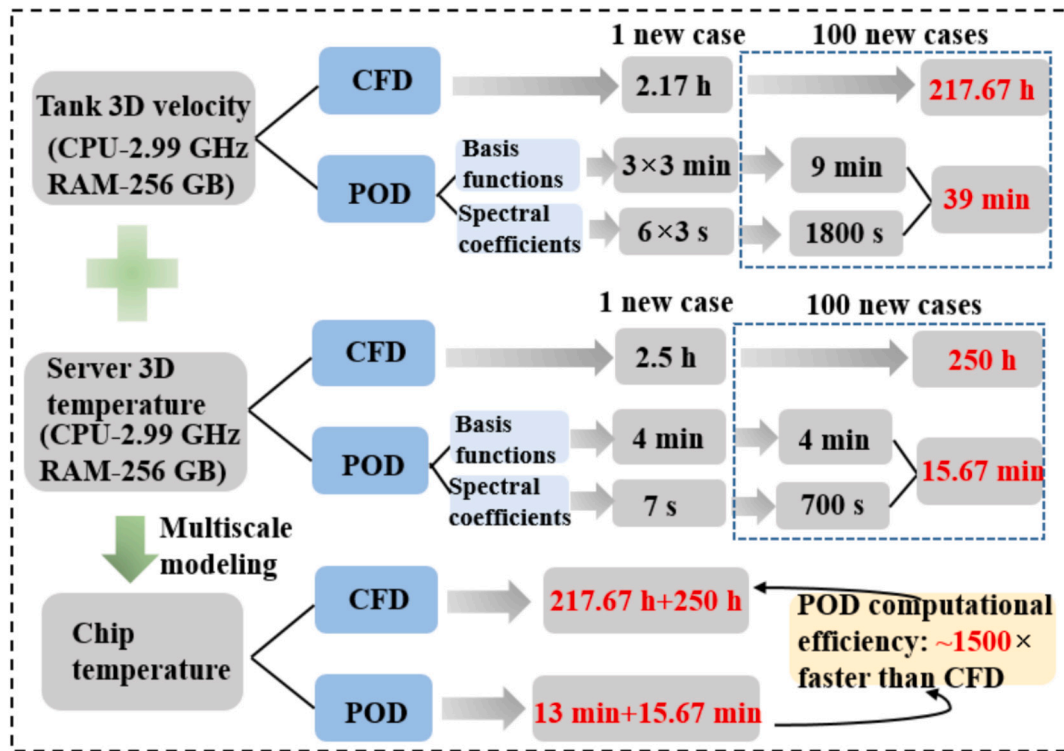


Fig. 16. Comparisons of computational time between POD-based ROM prediction and CFD simulation for the multiscale immersion cooling.

constructed POD-based ROM can accurately predict flow and thermal characteristics under various operating conditions. Specifically, compared with the CFD simulation results, the maximum average relative deviation of POD predictions is 2.88% for the inlet velocities while 2.92% for outlet velocities across all servers. The maximum average relative deviation for the full domain 3D velocity field prediction is 1.96%, and 3.11% for the temperature distribution at the server inlets. In terms of computational efficiency, the POD model achieves a speed up factor of approximately 333.3 times compared to CFD simulations for the velocity field, and around 1000 times for the temperature field.

- (2) When predicting the server chip temperature, the maximum average relative deviation of the 3D temperature field obtained using the two-scales POD-based ROM and the multiscale CFD simulation model is 3.7%. The POD model achieves approximately 1500 times improvement in computational efficiency compared with the full-order CFD simulation. Due to the complex variation in server power consumption and the non-uniform cooling conditions at different locations within the immersion tank, significant temperature differences arise among the server chips. Under the operating conditions of an inlet velocity of $1.48 \text{ m}\cdot\text{s}^{-1}$ and a total power consumption of 10.33 kW, the results reveals that the chip temperature of Server 3, as predicted by the two-scales POD model, reaches a maximum of 80.23°C among all studied scenarios. In contrast, the lowest chip temperature among the other servers under the same condition does not exceed 44.73°C , resulting in a maximum temperature difference of 35.5°C .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2026.130629>.

Data availability

Data will be made available on request.

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